

And Then, There is Embedded Vision!

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Last May, several leading technology firms got together in Oakland, USA, to form the Embedded Vision Alliance (EVA). The initiative's motto: To enable "machines that see!" This concept generally refers to machines that understand their environment through visual means.

Till April 25th this year, the Alliance had a membership of 20 companies. These included AMD, Analog Devices, Apical, Avnet Electronics Marketing, BDTI, CEVA, CogniMem, CogniVue, eyeSight, Freescale, IMS Research, Intel, MathWorks, Maxim Integrated Products, NVIDIA, National Instruments, Omek Interactive, Texas Instruments, Tokyo Electron Device and Xilinx.

First, what is embedded vision? It is said to be the merging of two technologies—embedded systems and computer vision (sometimes referred to as machine vision). An embedded system is any microprocessor-based system that isn't a general-purpose

Over the next few years, there will be a rapid proliferation of embedded vision technology into many kinds of embedded systems. Some of the key sectors worth looking up are retail, entertainment and medical

computer. Embedded systems are ubiquitous, and found in automobiles, kitchen appliances, consumer electronics devices, medical equipment, etc. Jeff Bier, president, Berkeley Design Technology (BDTI), and founder, EVA, is of the opinion that vision technologies could add valuable capabilities to electronic products.

Computer vision is the use of digital processing and intelligent algorithms to interpret meaning from images or video. Today, a major transformation is underway. Due to the emergence of very powerful, low-cost and energy-efficient processors, it has become possible to incorporate vision capabilities in a range of embedded systems.

EVA points to the example of the Microsoft Kinect video game controller that